

TOWARDS REVERSIBLE MODEL MERGING FOR LOW-RANK WEIGHTS

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ABSTRACT

Model merging seeks to consolidate a multiplicity of fine-tuned models into a single set of weights that generalizes competently across all constituent source tasks. Although the extant literature has established that merging can approximate the task-wise performance of the individual fine-tuned models from which it is derived, it has largely neglected the increasingly prevalent regime in which models are compressed into low-rank representations—whether through low-rank adaptation (LoRA) or through post-training singular value decomposition (SVD). We first demonstrate that the naive application of conventional merging procedures to low-rank weights precipitates a severe deterioration in the performance of the merged model. Motivated by this pathology, we advocate a fundamentally distinct formulation: rather than collapsing the full complement of adapters into one set of weights, we construct a compact basis (equivalent, for instance, to retaining two or more models) from which the original task-specific models can be recovered by means of linear combination. This reconceptualizes merging as the generation of a reconstruction-capable model space, in contradistinction to the production of a solitary merged model. Crucially, such a formulation permits us to “revert” to each individual model on demand, in recognition of the fact that no merged model can consistently surpass one specialized for its own task. Building upon this insight, we introduce our method, Reversible Model Merging (RMM), an efficient, data-free, and flexible procedure that furnishes a closed-form solution for jointly selecting the optimal basis of model weights and the task-specific coefficients governing their linear combination. Extensive experimentation spanning diverse datasets and model scales establishes that RMM consistently outperforms prevailing merging approaches, preserving the performance of low-rank compressed models by a substantial margin.

1 INTRODUCTION

Parameter-efficient fine-tuning methods, exemplified by Low-Rank Adapters (LoRA) (Hu et al., 2021), have established themselves as a cornerstone for the adaptation of large language models to a heterogeneous array of downstream tasks without incurring the prohibitive expense of full re-training. Concomitantly, model compression techniques—most notably post-training singular value decomposition (SVD) truncation—have been extensively adopted as a means of curtailing storage and inference costs, thereby furnishing an additional axis of efficiency (Ryu et al., 2023; Ping et al., 2024; Wang et al., 2025b;a). Alongside these developments, model merging has emerged as an auspicious paradigm for the consolidation of knowledge accrued across multiple task-specific models into a single unified model. Recent methodological advances, spanning the spectrum from elementary parameter interpolation to optimization-based procedures, have demonstrated that merged models can attain competitive performance across their source tasks without necessitating access to the original training data (Matena & Raffel, 2022; Yang et al., 2024; Yadav et al., 2023; Wei et al., 2025).

Notwithstanding this progress, a consequential limitation has remained conspicuously underexamined: even the most sophisticated merging strategies persistently fail to recover the performance of the original individual fine-tuned models. This performance shortfall becomes especially pronounced when the models subjected to merging have been compressed into low-rank form, whether through LoRA fine-tuning or through post-training SVD truncation. While merging endows a single

model with the capacity to address multiple tasks, it does so by irretrievably collapsing distinct task-specific representations into one shared parameter space—a compression that frequently engenders inter-task interference and, in the process, erodes the specialized adaptations painstakingly acquired by each constituent model (Gargiulo et al., 2025).

Although low-rank adapters and SVD-truncated models are individually compact, the unchecked proliferation of such compressed artifacts precipitates a new scalability bottleneck. In realistic multi-task, federated, and continual learning settings, it is commonplace to amass dozens—or indeed hundreds—of task-specific low-rank adapters, each of which demands separate storage as well as attendant coordination and management overhead. The retention of the entire collection erodes the very storage economies that low-rank parameterization was conceived to deliver, and correspondingly renders deployment more unwieldy. These considerations collectively motivate the development of merging strategies that operate directly upon low-rank representations—not merely to consolidate storage and abate redundancy, but equally to facilitate more efficient sharing, deployment, and cross-task transfer of compressed models.

In order to elucidate the limitations of prevailing approaches, we first interrogate the behavior of existing merging techniques as they are brought to bear upon low-rank compressed models. Our attention centers specifically on deltas—that is, the weight differences separating each fine-tuned model from the shared pre-trained initialization. Conventional merging strategies endeavor to amalgamate these deltas into a single set of parameters, predicated on an implicit assumption of linear compatibility. We demonstrate empirically, however, that the direct merging of low-rank deltas induces catastrophic performance degradation. This failure is attributable to two principal factors: the circumscribed expressive capacity intrinsic to low-rank representations, and the misalignment of task-specific subspaces—each optimized in isolation—which occasions severe interference once the subspaces are combined.

Prompted by this observation, we step back and pose a sequence of questions: *What if the objective of merging were not to produce a single terminal model? What if we were to relax the problem so as to permit the retention of more than one model? What if we could preserve the capacity to reconstruct the original models on demand?* Rather than collapsing a multitude of low-rank adapters into a single set of weights, we propose to maintain a compact set of weights that functions as a shared basis from which each original model may be recovered through linear combination. This reframing unlocks a hitherto underexplored design space: instead of compressing all information into a solitary model, we preserve the complementary representational directions distributed across a small ensemble of basis models, thereby enabling a post-merging “reversion” to the original adapters. Our approach thus contests the entrenched assumption that merging must culminate in a single model and, in its stead, redefines merging as the construction of a reconstruction-capable model space.

Building upon this reconstruction-driven perspective, we develop a theoretical framework that characterizes the optimal choice of basis for the minimization of reconstruction error. This development culminates in our method, Reversible Model Merging (RMM), which reinterprets the merging process through the lens of basis selection. Concretely, we formalize merging as a basis-selection problem in which the objective is to identify a compact set of task vectors that enables the accurate reconstruction of all original models. Within this framework, RMM furnishes a closed-form solution that simultaneously identifies an optimal set of basis task vectors and determines the task-specific coefficients whose linear combination recovers the individual task adaptations. We validate RMM empirically across a broad spectrum of datasets and model configurations, establishing that it consistently surpasses conventional merging strategies by a considerable margin. RMM attains this superiority while preserving the efficiency dividends of low-rank parameterization, thereby underscoring its practicality for large-scale multi-task adaptation. Moreover, it affords a flexible trade-off between storage and performance by way of a tunable hyperparameter that governs the number of final models retained.

2 RELATED WORK

Model merging has established itself as a potent strategy for consolidating the knowledge distributed across multiple fine-tuned models. Extant methods partition broadly into two categories: parameter-space manipulations that dispense with data altogether, and optimization-based approaches that cap-

italize upon data or auxiliary objectives. In what follows, we survey representative techniques drawn from both lines of inquiry:

Task Arithmetic: A foundational notion underpinning model merging is that of task arithmetic (Ilharco et al., 2023). Within this paradigm, a delta is construed as the difference between the parameters of a fine-tuned model and those of its pre-trained initialization. The cardinal insight animating task arithmetic is that such deltas comport themselves as modular building blocks: they may be added to a base model so as to confer novel abilities, subtracted so as to suppress undesired behaviors, or combined through scaling so as to blend a plurality of capabilities. This intuitive linear framework laid a robust foundation for the composition of model behaviors directly within weight space.

DARE: Although the naive addition of deltas can prove effective, it frequently breaks down when the merged models harbor conflicting or overlapping parameter updates. To redress this shortcoming, Yu et al. (2024) introduced DARE (Drop and Rescale), a technique predicated upon aggressive randomized pruning. The method proceeds by discarding the overwhelming majority of parameter changes—as much as 90-99%—on the premise that the preponderance of updates are inessential and may indeed interfere with the knowledge encoded in other models. The surviving updates are thereafter amplified so as to preserve the overall scale of the modification, thereby retaining the core task-specific modifications while attenuating interference.

TIES: In contradistinction to the randomized pruning of DARE, TIES-merging advances a deterministic framework for the mitigation of parameter conflicts (Yadav et al., 2023). It unfolds across three phases—Trim, Elect Sign, and Merge—each devised to resolve inter-model interference in a systematic fashion. In the first, only those parameters possessing the largest magnitudes are retained, so as to guarantee the persistence of the most influential fine-tuning updates. In the second, the method inspects, at each parameter position, the sign of the updates across models and enforces concordance by expunging any contribution that opposes the majority sign. In the third and final phase, the sign-consistent parameters are averaged to yield the merged representation. This consensus-driven procedure curtails destructive interference and produces a more stable merged model. Motivated by the same imperative of interference reduction that animates DARE and TIES, EMR-Merging (Elect, Mask and Rescale) proposes to elect a unified model and to affix task-specific binary masks and rescalers derived from each individual model, thereby steering the unified model toward each constituent task (Huang et al., 2024).

Other Merging Methods: Beyond purely parameter-space manipulations, a number of merging approaches are contingent upon explicit optimization procedures or upon direct access to some modest dataset. Methods such as model soups (Wortsman et al., 2022) refine the merging weights of task arithmetic by minimizing the loss incurred on validation data, thereby steering the interpolation toward superior generalization. In a kindred spirit, Fisher-weighted averaging (Matena & Raffel, 2022) incorporates second-order information to guide the merge, apportioning greater importance to those parameters estimated to be more critical for task performance. Yet other lines of inquiry cast merging explicitly as an optimization problem: RegMean (Jin et al., 2023) and AdaMerging (Yang et al., 2024), for instance, adaptively search for task-dependent merging coefficients by solving small-scale training objectives defined, respectively, over the discrepancy between the predictions of the merged and individual models and over the entropy of predictions on unlabeled data. A more recent formulation reduces model merging to the minimization of the loss difference between the merged model and the individual fine-tuned models, invoking a first-order Taylor expansion so as to derive a tractable surrogate objective (Wei et al., 2025). While such approaches can elicit stronger task-specific performance, they typically presuppose access to labeled or unlabeled data, supplementary optimization, or both—requirements that circumscribe their applicability in regimes where merging must proceed in the absence of retraining and data.

3 PRELIMINARIES

We address the problem of consolidating a collection of fine-tuned models into a single unified model by means of model merging. Let $\{\theta_1, \dots, \theta_n\}$ denote n models fine-tuned on n distinct tasks. A merging algorithm $\mathcal{M}$ combines these models into a single set of weights $\theta_{\mathcal{M}} = \mathcal{M}(\{\theta_i\}_{i=1}^n)$. The objective is to identify a merging procedure $\mathcal{M}$ that maximizes the average validation perfor-

mance attained across the full complement of tasks:

$$\max_{\mathcal{M}} \frac{1}{n} \sum_{i=1}^n \text{Perf}(\theta_{\mathcal{M}}, \mathcal{D}_i), \quad (1)$$

where $\mathcal{D}_i$ denotes a held-out validation set for task i , and Perf denotes any suitable performance metric (e.g., accuracy or F1 score). It is conventional to posit that all fine-tuned models emanate from a common pretrained initialization θ_{pre} . For any layer l , the update (or delta) of model i relative to the base model is defined as $\Delta_i^l = \theta_i^l - \theta_{\text{pre}}^l$, wherein $\Delta_i^l \in \mathbb{R}^{m \times d}$ designates the weight difference for a layer of dimensions $m \times d$.

3.1 LOW-RANK COMPRESSED MODELS

In practice, it is customary to compress fine-tuned models into low-rank form in order to curtail storage, training, and inference costs. Two compression strategies enjoy widespread adoption:

LoRA fine-tuning: In the course of training, each layer is augmented with trainable low-rank matrices $A \in \mathbb{R}^{m \times r}$ and $B \in \mathbb{R}^{r \times d}$, where $r \ll \min(m, d)$. The pretrained weights are held frozen throughout, and only A and B are subjected to optimization.

Post-training SVD truncation (PT-SVD): Subsequent to standard fine-tuning, the update Δ_i^l is decomposed via SVD and truncated so as to retain only its top- r singular components, thereby yielding a low-rank approximation $\hat{\Delta}_i^l = A_i^l B_i^l$, for $A_i^l \in \mathbb{R}^{m \times r}$ and $B_i^l \in \mathbb{R}^{r \times d}$ where $r \ll \min(m, d)$.

In either case, the fine-tuned update for layer l in model i admits a low-rank representation $\hat{\Delta}_i^l = A_i^l B_i^l$, in which $A_i^l \in \mathbb{R}^{m \times r}$ and $B_i^l \in \mathbb{R}^{r \times d}$, with r denoting the rank of compression. The merging problem thereby reduces to one of combining low-rank updates across tasks. The goal is to construct a merged model that integrates the compressed deltas $\hat{\Delta}_i = \{\hat{\Delta}_i^l\}_{l=1}^L$ across the totality of tasks:

$$\hat{\theta}_{\mathcal{M}} = \theta_{\text{pre}} + \mathcal{M}(\{\hat{\Delta}_i\}_{i=1}^n), \quad \max_{\mathcal{M}} \frac{1}{n} \sum_{i=1}^n \text{Perf}(\hat{\theta}_{\mathcal{M}}, \mathcal{D}_i), \quad (2)$$

where L denotes the total number of layers. Throughout the present work, we concentrate on precisely this setting, in which each model is compressed into low-rank form. The section that follows investigates how conventional merging strategies comport themselves under the low-rank regime.

4 MOTIVATING OBSERVATION

We now scrutinize how conventional model merging strategies behave once applied to the low-rank compressed models introduced in the preceding section. For each layer l , two natural extensions of merging have been entertained in prior work (Tang et al., 2025; Zhao et al., 2025; Zheng et al., 2025; Zhang et al., 2025):

- **Combined merging:**

$$\hat{\theta}_{\mathcal{M}}^l = \theta_{\text{pre}}^l + \mathcal{M}(\{A_i^l B_i^l\}_{i=1}^n),$$

wherein each low-rank update is first reconstituted into its full-rank form prior to being merged.

- **Separate merging:**

$$\hat{\theta}_{\mathcal{M}}^l = \theta_{\text{pre}}^l + \mathcal{M}(\{A_i^l\}_{i=1}^n) \mathcal{M}(\{B_i^l\}_{i=1}^n),$$

wherein the low-rank factors are merged independently and only thereafter recomposed.

While both approaches are theoretically admissible, the combined strategy proves acutely inefficient in practice. Specifically, storing the merged update in its combined form necessitates retaining an $m \times d$ matrix, whereas the separate approach retains no more than the equivalent of one of the low-rank factors, $A_i^l \in \mathbb{R}^{m \times r}$ and $B_i^l \in \mathbb{R}^{r \times d}$. Since $r(m + d) \ll md$ whenever $r \ll \min(m, d)$, the storage and memory overhead entailed by the combined approach subverts the very rationale for

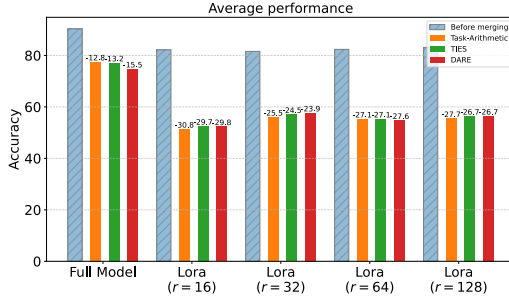


Figure 1: Average task performance of the individual models (prior to merging) juxtaposed against that of the merged models, reported across distinct merging methods. Results are presented both for fully fine-tuned models and for models fine-tuned by means of LoRA at a range of ranks $r \in \{16, 32, 64, 128\}$.

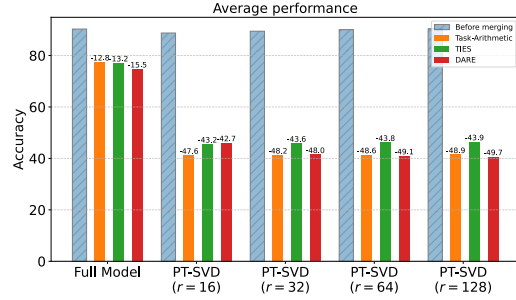


Figure 2: Average task performance of the individual models (prior to merging) juxtaposed against that of the merged models, reported across distinct merging methods. Results are presented both for fully fine-tuned models and for models compressed by means of PT-SVD at a range of ranks $r \in \{16, 32, 64, 128\}$.

compression. By way of illustration, consider a layer with $m = d = 1024$ and a compression rank $r = 16$. The combined approach must store a full update comprising $md = 1,048,576$ parameters, whereas the separate approach need store only $r(m+d) = 16 \times (1024+1024) = 32,768$ parameters.

This amounts to a **32 $\times$ reduction in storage**, thereby demonstrating that the combined approach comprehensively negates the efficiency benefits conferred by low-rank compression. We therefore adopt the separate strategy as the sole viable option for merging compressed models in practice.

We applied a number of established model merging baselines to low-rank compressed models across varying compression ratios, invoking throughout the separate approach delineated above. RoBERTa-base (Zhuang et al., 2021) was designated as the pre-trained model, and experiments were conducted on 5 GLUE tasks (Wang et al., 2018) (further details are deferred to Appendix A.3). As Figures 1 and 2 make manifest, the merging of compressed models precipitates a *catastrophic drop* in average task performance relative to the merging of full-rank models. We ascribe this degradation to two concomitant factors: (i) **Amplified task interference**: compression relinquishes much of the representational capacity, rendering low-rank models markedly more susceptible to destructive interference upon merging; (ii) **Loss of coupling between A and B** : conventional merging treats the low-rank matrices A_i^l and B_i^l as independent factors, disregarding the intrinsically coupled nature of their product and thereby introducing structural misalignment into the merged updates. Acting in concert, these factors severely impair the performance of merged low-rank models and intimate a more profound incompatibility between traditional merging methods and compressed representations.

This observation crystallizes into a pivotal insight: under conventional strategies, no merged model can reliably recuperate the task-specific performance of the original low-rank adapters. Such a state of affairs prompts a natural question: *Can we instead reconstruct the original compressed models, or approximate them closely, by retaining a compact set of basis models—perhaps equivalent to two or more adapters, contingent upon our storage budget?* Rather than coercing all updates into a single representation, we explore an alternative design space in which merging constructs a *reconstruction-capable model space*. In what follows, we formalize this perspective and introduce our proposed method, *Reversible Model Merging* (RMM).

5 REVERSIBLE MODEL MERGING

The empirical failures of conventional merging in the compressed setting are symptomatic of a fundamental limitation: no single merged model can faithfully preserve the task-specific performance of all low-rank compressed models simultaneously. Prompted by this diagnosis, we propose a new approach, RMM, that forsakes the objective of collapsing all models into a single set of weights. In

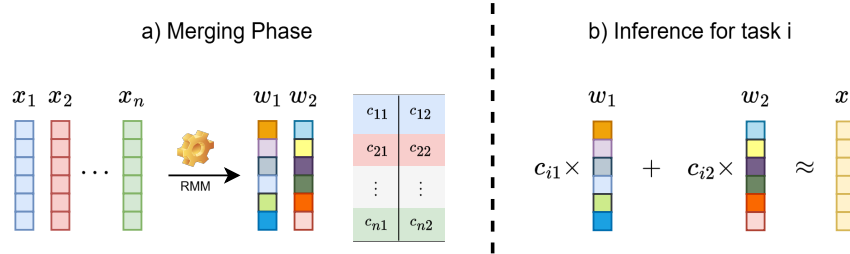


Figure 3: Overview of the proposed RMM framework. **Left:** Over the course of the merging phase, we compute a compact set of basis directions (two, in the present example) together with their associated task-specific coefficients. **Right:** At inference time, a target model i is reconstructed by linearly combining the basis vectors under the corresponding coefficients.

its place, RMM constructs a diminutive set of shared components—basis models—from which all original task models can be reconstructed to a close approximation.

Rather than coercing all n task-specific models into a single merged solution $\hat{\theta}_{\mathcal{M}}$, RMM retains a compact set of p merged components, where $2 \leq p \ll n$. These components span a low-dimensional subspace that captures the essential variations manifest across tasks, thereby enabling the efficient recovery of each model on demand. The design deliberately balances performance against memory: by tuning p , practitioners may negotiate a trade-off between storage cost and reconstruction accuracy.

For each layer l , we direct our attention to the r -dimensional vectors residing within the low-rank delta matrices (A_i^l and B_i^l)—namely, the rows of A_i^l and the columns of B_i^l (r being the compression rank of the compressed models). We construe each row of A_i^l and each column of B_i^l as an independent *task vector*, and we designate a specific row or column index as the *task vector position*. For any fixed position (i.e., a given row or column index), we assemble the corresponding task vectors across all n models, denoting the task vector contributed by model i at that position as $x_i \in \mathbb{R}^r$. Our algorithm is subsequently applied independently to each such position (whether a row index in A_i^l or a column index in B_i^l): we compress the set of task vectors $\{x_i\}_{i=1}^n$ by projecting them onto a shared basis, so as to permit their later reconstruction as required. There are, in aggregate, $m + d$ task vector positions per layer— m rows drawn from A_i^l and d columns drawn from B_i^l —and the algorithm is applied separately to each of them.

To this end, we seek a set of basis vectors $\{w_1, \dots, w_p\}$, wherein each $w_j \in \mathbb{R}^r$ for $1 \leq j \leq p$. The span of these basis vectors delimits a subspace $W = [w_1 \dots w_p] \in \mathbb{R}^{r \times p}$ onto which each task vector x_i is projected, subject to the requirement that the projection reconstruct x_i with the utmost fidelity. We further introduce $C \in \mathbb{R}^{n \times p}$ as the coefficient matrix, whose i -th row $[c_{i1}, \dots, c_{ip}]$ enumerates the scalar weights by which the basis vectors are linearly combined in reconstructing x_i . The goal, accordingly, is to determine W and C so as to minimize the aggregate reconstruction error incurred across all n task vectors. This desideratum gives rise to the following optimization problem:

$$\min_{W, C} \sum_{i=1}^n \left\| x_i - \sum_{j=1}^p c_{ij} w_j \right\|_2^2. \quad (3)$$

This formulation entails storing rp parameters for the basis and np for the coefficients, for a total of $p(r + n)$ —a substantial reduction relative to the rn parameters that would be required to store all task vectors outright. Of particular note, the value of p constitutes a tunable hyperparameter that arbitrates the performance-storage trade-off. Figure 3 illustrates our approach for the case $p = 2$. The optimization in equation 3 may be recast in matrix form as follows:

$$\min_{W, C} \|X - CW^\top\|_F^2, \quad (4)$$

where $X = [x_1 \dots x_n]^\top \in \mathbb{R}^{n \times r}$ denotes the matrix aggregating all task vectors. It is well established that, for any given W , the optimal C is furnished by the least-squares solution $C^* = XW(W^\top W)^{-1}$. Upon stipulating that the basis vectors be orthonormal ($W^\top W = I_p, p < r$),

the optimization admits the reformulation:

$$\min_{\mathbf{W}} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 \quad \text{s.t.} \quad \mathbf{W}^\top\mathbf{W} = \mathbf{I}_p. \quad (5)$$

Observe that, within this formulation, we shall store $\mathbf{W} \in \mathbb{R}^{r \times p}$ together with $\mathbf{X}\mathbf{W} \in \mathbb{R}^{n \times p}$ (the product $\mathbf{X}\mathbf{W}$ being equivalent to $\mathbf{C}$ in equation 4).

Theorem 1 (Optimal Solution to equation 5). *Let $\mathbf{X} \in \mathbb{R}^{n \times r}$ be zero-centered and $p < \min\{r, n\}$. The optimal basis $\mathbf{W}^*$ that minimizes equation 5 is given by the top- p eigenvectors of the sample covariance matrix $\mathbf{X}^\top\mathbf{X}$. Equivalently, $\mathbf{W}^*$ consists of the top- p right singular vectors of $\mathbf{X}$.*

The proof of Theorem 1 is deferred to Appendix A.1. We now proceed to delineate the two constituent phases of RMM.

5.1 MERGING PHASE

We construct a compact representation of all task-specific models by recourse to a shared basis. For each layer and for each task vector position (a fixed row in $\mathbf{A}_i^l$ or column in $\mathbf{B}_i^l$), we assemble the task vectors across all n models as $\mathbf{X} = [\mathbf{x}_1 \dots \mathbf{x}_n]^\top \in \mathbb{R}^{n \times r}$. We first compute the mean vector $\boldsymbol{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$ and thereupon zero-center $\mathbf{X}$ (i.e., $\mathbf{X} \leftarrow [\mathbf{x}_1 - \boldsymbol{\mu} \dots \mathbf{x}_n - \boldsymbol{\mu}]^\top$). Subsequently, we perform an SVD of the centered matrix, $\mathbf{X} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$, and retain its top- p right singular vectors so as to constitute the optimal orthonormal basis $\mathbf{W}^*$ that minimizes the reconstruction error in equation 5, i.e., $\mathbf{W}^* = \mathbf{V}_{[:,1:p]} \in \mathbb{R}^{r \times p}$. We thereafter compute the corresponding coefficient matrix $\mathbf{C}^* = \mathbf{X}\mathbf{W}^* \in \mathbb{R}^{n \times p}$, which stores the linear-combination weights requisite for reconstructing the original task vectors. The i -th row of $\mathbf{C}^*$, $[c_{i1}, \dots, c_{ip}]$, contains precisely the coefficients needed to reconstruct $\mathbf{x}_i$.

At the conclusion of this phase, the original task vectors $\{\mathbf{x}_i\}_{i=1}^n$ have been discarded, and only the basis $\mathbf{W}^*$, the coefficients $\mathbf{C}^*$, and the mean vector $\boldsymbol{\mu}$ are retained for each task vector position. In aggregate, this representation effects a memory-efficient compression from rn parameters (required to retain all task vectors) down to $p(r+n) + r$ (per task vector position). We remark that the merging phase is executed offline, in advance of inference.

5.2 INFERENCE PHASE

In keeping with common practice in model merging, we assume that the target task associated with each input is either explicitly stipulated by the user or automatically inferred by an oracle router that assigns the input to the most apposite model index. Once the target task index i has been established, our objective is to reconstruct its task vector $\mathbf{x}_i$ —denoted $\hat{\mathbf{x}}_i$ —by leveraging the compact representation $(\mathbf{W}^*, \mathbf{C}^*, \boldsymbol{\mu})$ obtained during the merging phase at each task vector position. We thereupon recover $\mathbf{A}_i^l$ and $\mathbf{B}_i^l$ by concatenating the reconstructed task vectors (the $\hat{\mathbf{x}}_i$ vectors) across all positions (i.e., across all the pertinent rows of $\mathbf{A}_i^l$ or columns of $\mathbf{B}_i^l$).

Recall that, for each task vector position, $\mathbf{W}^* \in \mathbb{R}^{r \times p}$ encodes the orthonormal basis vectors, $\mathbf{C}^* \in \mathbb{R}^{n \times p}$ stores the task-specific coefficients, and $\boldsymbol{\mu} \in \mathbb{R}^r$ constitutes the mean vector across all task vectors. The reconstruction is effected by linearly combining the basis directions under the task-specific coefficients and subsequently shifting by the mean vector $\boldsymbol{\mu}$, as follows:

$$\hat{\mathbf{x}}_i = \sum_{j=1}^p c_{ij}^* \mathbf{w}_j^* + \boldsymbol{\mu} = \mathbf{C}_{[i,:]}^* \mathbf{W}^{*\top} + \boldsymbol{\mu}.$$

The resulting $\hat{\mathbf{x}}_i \in \mathbb{R}^r$ thus furnishes an approximation of the original task vector $\mathbf{x}_i$ that had been discarded during the merging phase. This reconstruction is carried out independently for each task vector position (every row of $\mathbf{A}_i^l$ and every column of $\mathbf{B}_i^l$) and across all layers, thereby reconstituting in full the low-rank delta for task-specific model i , namely $\hat{\Delta}_i$. The complete procedure spanning both the merging and the inference phases is set out in Algorithm 1. We next present an empirical evaluation of our method and juxtapose its performance against that of standard baselines.

Algorithm 1 Reversible Model Merging (RMM)

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1: Merging Phase:
2: Require: Low-rank deltas  $A_i^l$  and  $B_i^l$ , compression rank  $r$ , number of basis vectors  $p$ 
3: for  $l = 1$  to  $L$  do
4:   for each task vector position (each row in  $A_i^l$  or each column in  $B_i^l$ ) do
5:     Gather task vectors  $x_i$  of each model  $i$  ( $1 \leq i \leq n$ )
6:      $X \leftarrow [x_1 \dots x_n]^\top \in \mathbb{R}^{n \times r}$ 
7:      $\mu \leftarrow \sum_{i=1}^n x_i \in \mathbb{R}^r$ 
8:      $X \leftarrow [x_1 - \mu \dots x_n - \mu]^\top$ 
9:      $U, \Sigma, V^\top \leftarrow \text{SVD}(X)$ 
10:     $W^* \leftarrow V_{[:,1:p]} \in \mathbb{R}^{r \times p}$ 
11:     $C^* \leftarrow XW^* \in \mathbb{R}^{n \times p}$ 
12:    Store  $(W^*, C^*, \mu)$  and discard task vectors  $\{x_1 \dots x_n\}$ 
13:   end for
14: end for
15: Inference Phase:
16: Require: target task  $i$ , stored representation  $(W^*, C^*, \mu)$  for each task vector position
17: for  $l = 1$  to  $L$  do
18:   for each task vector position do
19:      $\hat{x}_i = C_{[i,:]}^* W^{*\top} + \mu$ 
20:   end for
21:    $\hat{A}_i^l = \text{Concatenate}(\hat{x}_i \text{ vectors across all row positions})$ 
22:    $\hat{B}_i^l = \text{Concatenate}(\hat{x}_i \text{ vectors across all column positions})$ 
23: end for
24:  $\hat{\Delta}_i = \{\hat{A}_i^l \hat{B}_i^l\}_{l=1}^L$ 
25: Inference with the reconstructed model  $\theta_{pre} + \hat{\Delta}_i$ 
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6 EXPERIMENTS

In order to appraise the efficacy of RMM, we undertake extensive experimentation spanning a diversity of compression settings, tasks, and model architectures. Our aim is twofold: to ascertain whether RMM can preserve the performance of low-rank task-specific models while simultaneously curtailing the overall storage cost, and to establish how it fares in comparison with conventional merging baselines.

6.1 EXPERIMENTAL SETUP

We first evaluate RMM on eight RoBERTa-base models fine-tuned across eight distinct tasks (Zhuang et al., 2021). Each model is individually compressed by means of either PT-SVD or LoRA at target ranks $r \in \{16, 32, 64, 128\}$. Following compression, we apply RMM to the resulting low-rank models and contrast its performance with that of standard merging baselines—Task Arithmetic (TA) (Ilharco et al., 2023), TIES-merging (Yadav et al., 2023), and DARE (Yu et al., 2024). As an upper bound, we additionally report the performance of the individual compressed models prior to merging. Supplementary results are furnished in Appendix A.2, encompassing experiments on OPT-1.3b (Zhang et al., 2022) as well as on ViT-B/32 (Radford et al., 2021) for vision tasks, under differing numbers of tasks.

In conformity with the GLUE benchmark protocol (Wang et al., 2018), we evaluate each task according to its standard metric. QNLI (question–answering inference), MRPC (paraphrase detection), SST-2 (sentiment classification), MNLI (multi-genre natural language inference), QQP (duplicate question detection), and RTE (textual entailment) are each quantified by accuracy; CoLA (linguistic acceptability) is assessed via Matthews correlation, whereas STS-B (semantic textual similarity) is gauged by Pearson correlation. As an overall performance measure, we report the average score computed across all tasks. We further quantify, for each method, the relative storage cost—defined as the fraction of parameters retained relative to the storage of all individual task-specific adapters. Baseline methods, which store but a single merged model, incur a fixed cost of $\frac{r}{rn} = \frac{1}{n}$. For RMM

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	RTE	COLA	STS-B	Average
16	No merging	100%	92.02	85.68	93.69	85.21	87.19	77.98	58.05	90.59	83.80
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.19	31.66
	TIES	13%	50.56	34.78	49.08	31.82	36.82	47.29	0.00	3.75	31.76
	DARE	13%	50.80	33.51	49.08	31.82	36.82	47.29	0.00	5.07	31.80
	RMM ($p = 2$)	50%	51.20	78.20	88.30	66.74	86.02	47.29	6.56	33.59	57.24
	RMM ($p = 3$)	69%	88.83	86.26	92.89	79.58	87.18	48.74	33.85	60.46	72.22
32	No merging	100%	92.31	85.86	94.15	86.38	88.62	78.34	59.33	90.68	84.46
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.21	31.66
	TIES	13%	50.56	35.19	49.08	31.82	36.82	47.29	0.00	3.87	31.83
	DARE	13%	50.72	33.51	49.08	31.82	36.82	47.29	0.00	3.48	31.59
	RMM ($p = 2$)	44%	50.54	74.32	55.05	62.68	87.07	47.29	0.00	23.18	50.02
	RMM ($p = 3$)	59%	84.66	86.32	92.89	79.70	88.05	47.29	23.57	44.54	68.38
64	No merging	100%	92.51	86.03	94.38	87.18	90.15	78.70	61.11	90.72	85.10
	TA	13%	50.61	33.51	49.08	31.82	36.82	47.29	0.00	4.23	31.67
	TIES	13%	50.58	35.13	49.08	31.82	36.82	47.29	0.00	3.80	31.82
	DARE	13%	50.58	33.51	49.08	31.82	36.82	47.29	0.00	5.22	31.79
	RMM ($p = 2$)	41%	50.53	73.10	49.08	58.48	86.98	47.29	0.00	17.50	47.87
	RMM ($p = 3$)	55%	74.14	86.20	92.43	78.83	89.00	47.29	13.16	33.70	64.34
128	No merging	100%	92.68	86.14	94.50	87.33	91.05	79.78	60.16	90.74	85.30
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.23	31.67
	TIES	13%	50.58	35.54	49.08	31.82	36.82	47.29	0.00	3.93	31.88
	DARE	13%	50.61	34.78	49.08	31.82	36.82	47.29	0.00	2.41	31.60
	RMM ($p = 2$)	39%	50.54	70.78	49.08	54.69	84.68	47.29	0.00	14.53	46.45
	RMM ($p = 3$)	52%	59.20	85.80	92.32	77.50	89.02	47.29	6.56	26.94	60.58

Table 1: Performance on the GLUE benchmark for the merging of eight RoBERTa-base models compressed via PT-SVD across a range of ranks ($r = 16, 32, 64, 128$).

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	RTE	COLA	STS-B	Average
16	No merging	100%	61.38	83.48	94.50	83.90	87.56	68.23	53.92	88.71	77.71
	TA	13%	50.59	33.51	72.13	34.79	64.46	47.29	0.00	21.81	40.57
	TIES	13%	50.63	33.68	74.31	35.64	67.45	47.29	0.00	26.38	41.92
	DARE	13%	50.63	33.51	73.74	35.97	65.35	47.29	0.00	21.62	41.01
	RMM ($p = 2$)	50%	51.13	34.14	90.71	60.48	84.41	47.29	0.00	43.14	51.41
	RMM ($p = 3$)	69%	55.92	38.14	93.12	82.95	87.30	47.29	6.56	57.81	58.64
32	No merging	100%	60.13	81.28	93.81	84.46	88.18	62.82	53.12	88.53	76.54
	TA	13%	50.91	68.06	54.70	33.77	65.50	55.96	0.00	-0.48	41.05
	TIES	13%	51.03	68.06	56.77	34.24	66.27	58.12	0.00	1.02	41.94
	DARE	13%	50.96	64.64	63.07	35.12	63.66	55.60	0.00	1.01	41.76
	RMM ($p = 2$)	44%	52.08	65.39	81.54	62.46	77.06	57.40	0.00	9.95	50.73
	RMM ($p = 3$)	59%	54.62	69.97	92.20	81.29	85.97	57.40	4.64	24.08	58.77
64	No merging	100%	59.62	84.12	94.15	85.20	88.75	67.15	58.31	88.70	78.25
	TA	13%	54.75	68.52	77.29	37.24	36.82	47.29	0.00	-3.53	39.80
	TIES	13%	54.82	68.75	80.73	37.67	36.82	47.29	0.00	-2.48	40.45
	DARE	13%	53.71	68.58	78.56	38.33	36.82	47.29	0.00	-4.28	39.88
	RMM ($p = 2$)	41%	55.74	69.28	87.27	42.81	77.72	47.29	0.00	4.91	48.13
	RMM ($p = 3$)	55%	59.07	70.14	90.83	79.11	82.06	47.29	0.00	14.70	55.40
128	No merging	100%	61.58	85.04	93.58	85.71	89.68	57.04	59.35	89.68	77.71
	TA	13%	51.49	69.22	54.47	37.16	63.27	47.29	0.00	-1.74	40.15
	TIES	13%	51.91	68.93	57.34	37.81	63.36	47.29	0.00	-0.25	40.80
	DARE	13%	51.97	68.99	58.60	35.64	63.33	47.29	0.00	-0.44	40.67
	RMM ($p = 2$)	39%	57.18	67.83	88.53	49.54	84.64	47.29	0.00	28.00	52.88
	RMM ($p = 3$)	52%	59.13	39.19	92.09	80.08	86.39	47.29	15.45	71.18	61.35

Table 2: Performance on the GLUE benchmark for the merging of eight RoBERTa-base models compressed via LoRA across a range of ranks ($r = 16, 32, 64, 128$).

endowed with p basis components, the relative storage cost is given by $\frac{p(r+n)+r}{rn}$, as derived in Section 5.1.

6.2 RESULTS AND DISCUSSION

Tables 1 and 2 collate the performance of RMM against that of the baselines under PT-SVD and LoRA compression, respectively. Across both regimes and for every target rank $r \in \{16, 32, 64, 128\}$, our method consistently outstrips TA, TIES, and DARE by a pronounced margin. At the lower ranks ($r = 16, 32$), the baseline methods fail catastrophically, attaining a mere

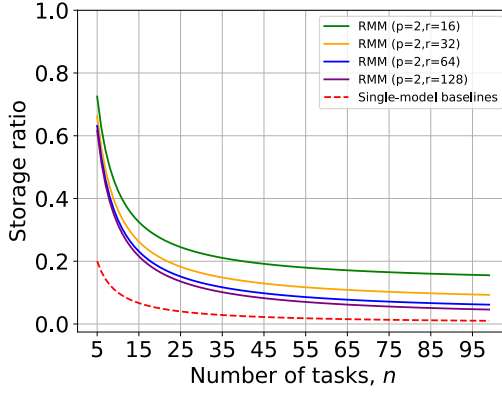


Figure 4: Scalability of RMM ($p = 2$). The storage ratio denotes the percentage of memory required relative to the storage of all models.

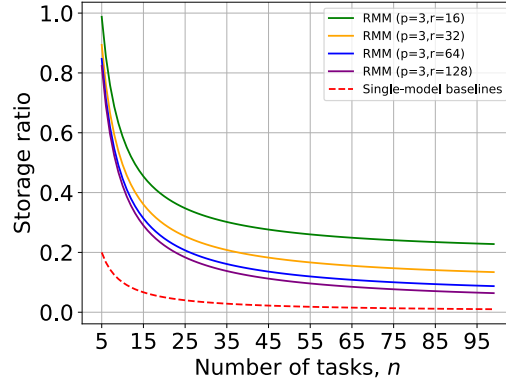


Figure 5: Scalability of RMM ($p = 3$). The storage ratio denotes the percentage of memory required relative to the storage of all models.

31–42% average score and faring especially poorly on structurally sensitive tasks such as CoLA and STS-B. By contrast, RMM with $p = 2$ realizes substantial gains (for instance, 57.24% under PT-SVD and 51.41% under LoRA at $r = 16$), while RMM with $p = 3$ yields stronger results still (reaching as high as 72.22% and 58.64% at $r = 16$), thereby appreciably narrowing the gap to the individual compressed models. At the higher ranks ($r = 64, 128$), where the compression error is attenuated, RMM continues to prevail over the baselines: with $p = 3$ it achieves 64.34% and 60.58% under PT-SVD and 55.40% and 61.35% under LoRA, consistently eclipsing the single-model baselines, which remain languishing near 40%. Even RMM with $p = 2$ retains a manifest advantage, corroborating that the reversible approach forestalls the destructive interference that so degrades the performance of prior methods.

We further observe that, as the rank increases, the relative advantage of RMM over the baselines progressively diminishes. One plausible account of this trend is that higher-rank task representations inhabit higher-dimensional spaces, whereupon they become more expressive yet less mutually aligned, rendering their compression into a shared low-dimensional basis correspondingly more arduous. Taken in aggregate, the results demonstrate that RMM confers a consistent advantage across both compression regimes and all ranks: it preserves high task performance at the expense of some modest additional storage, thereby furnishing a tunable trade-off between accuracy and storage efficiency through adjustment of the number of reversible components. Crucially, this storage cost is borne by **offline memory**—typically CPU-resident and inexpensive—whereas task performance bears directly upon runtime quality. In such circumstances, the slight augmentation of CPU-side storage frequently constitutes a worthwhile concession in exchange for markedly superior accuracy.

6.3 SCALABILITY OF RMM

We next interrogate the scalability of RMM as the number of models, n , grows. Figures 4 and 5 depict the storage ratio (relative to the storage of all individual task-specific models) as a function of the number of tasks, for RMM with $p = 2$ and $p = 3$, respectively. The results affirm that RMM scales favorably: as the number of tasks increases, the relative storage cost declines substantially. This trend throws into relief the cardinal benefit of reversible merging—namely, that storage grows sublinearly with the number of task-specific models. Such sublinear growth is a direct corollary of the RMM design: once the basis has been fixed, the incorporation of a new task necessitates no more than the storage of an additional row in the coefficient matrix. Traditional approaches, by contrast, must either store a single merged model (thereby sacrificing performance) or store each task-specific model in isolation (thereby scaling linearly). These findings underscore the practical utility of RMM in large-scale multi-task, continual, and federated settings, wherein it may prove necessary to administer hundreds of models efficiently without any concomitant compromise of task performance.

7 CONCLUSION

In the present work, we have introduced a principled framework for reversible model merging that reconceives the manner in which multiple fine-tuned models may be stored, merged, and reconstructed. Rather than compelling all task-specific knowledge into a single merged model, we relax this constraint by retaining a compact set of intermediate representations—substantially fewer than the total number of original models, yet appreciably more flexible than a solitary aggregate. This design enables the accurate reconstruction of task-specific models while effecting a significant reduction in storage overhead. Our method thereby affords a controllable trade-off between performance and storage efficiency, particularly within the context of low-rank compressed models. By representing fine-tuned models through compact bases together with reversible mappings, we demonstrate the feasibility of merging, reconstructing, and deploying models at high accuracy while adhering to stringent memory budgets—a flexibility that empowers practitioners to calibrate the degree of compression and the fidelity of reconstruction to the exigencies of their particular application.

Taken as a whole, our framework transcends heuristic or one-shot merging techniques by formalizing model merging as a structured and reversible process. This novel perspective not only enlarges the design space of efficient multi-task deployment but likewise opens auspicious avenues for future inquiry into performance-aware compression and reconstruction strategies.

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A APPENDIX

A.1 PROOF OF THEOREM 1

Let $\mathbf{A} = \mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top$. By the definition of the Frobenius norm, $\|\mathbf{A}\|_F^2 = \text{Tr}(\mathbf{A}^\top \mathbf{A})$, wherein $\text{Tr}(\cdot)$ denotes the trace of a square matrix. Expansion yields:

$$\|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 = \text{Tr}((\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)^\top (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)).$$

We proceed to expand the argument of $\text{Tr}(\cdot)$:

$$\begin{aligned} (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)^\top (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top) &= (\mathbf{X}^\top - \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top)(\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top) \\ &= \mathbf{X}^\top \mathbf{X} - \mathbf{X}^\top \mathbf{X}\mathbf{W}\mathbf{W}^\top - \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X} \\ &\quad + \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}\mathbf{W}^\top. \end{aligned}$$

We now take the trace of the expanded form. Invoking the cyclicity of the trace together with the orthonormality constraint $\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p$:

$$\begin{aligned} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - 2\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) + \text{Tr}(\mathbf{W}^\top \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) \\ &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - 2\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) + \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) \\ &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}). \end{aligned}$$

Since $\text{Tr}(\mathbf{X}^\top \mathbf{X})$ is a constant, the minimization of $\|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2$ is tantamount to the maximization of $\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W})$. Consequently,

$$\min_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 \iff \max_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p} \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}).$$

By the Rayleigh quotient theorem (Horn & Johnson, 2012), the maximizer is procured by taking the columns of $\mathbf{W}$ to be the top- p eigenvectors of $\mathbf{X}^\top \mathbf{X}$. This coincides precisely with the principal component analysis (PCA) solution: it isolates the p -dimensional subspace that maximizes the projected variance and, equivalently, minimizes the squared reconstruction error. Moreover, if $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ denotes the SVD of $\mathbf{X}$, then $\mathbf{X}^\top \mathbf{X} = \mathbf{V}\mathbf{\Sigma}^2\mathbf{V}^\top$, whence the eigenvectors of $\mathbf{X}^\top \mathbf{X}$ are precisely the right singular vectors (the columns of $\mathbf{V}$). It follows that the PCA solution may likewise be obtained by taking the top- p right singular vectors of $\mathbf{X}$. In practice, this SVD-based route is numerically more stable and frequently computationally preferable to the direct formation of $\mathbf{X}^\top \mathbf{X}$, particularly when r is large.

A.2 ADDITIONAL RESULTS

In this section, we present extended results for two supplementary pre-trained models: OPT-1.3b and ViT-B/32. For **OPT-1.3b**, we evaluate on six natural language understanding tasks drawn from GLUE and cognate benchmarks: QNLI, MRPC, SST-2, MNLI, QQP, and BoolQ (Clark et al., 2019). For **ViT-B/32**, we concentrate on vision classification tasks, spanning seven heterogeneous datasets: MNIST (LeCun et al., 1998), GTSRB (Stallkamp et al., 2011), SVHN (Netzer et al., 2011), Cars (Krause et al., 2013), EuroSAT (Helber et al., 2019), DTD (Cimpoi et al., 2014), and SUN397 (Xiao et al., 2010).

Table 3 reports the results for six OPT-1.3b models compressed via PT-SVD. Both variants of RMM ($p = 2$ and $p = 3$) consistently surpass the baselines, with RMM at $p = 3$ narrowing the gap to the original individual models while requiring substantially fewer parameters. Table 4 sets out the corresponding results for seven ViT-B/32 models. On vision tasks, the benefit conferred by RMM proves even more pronounced: with $p = 3$, our method attains an average accuracy less than 3% below that of the original models, all the while reducing storage to a mere 59% of that required to retain all models at $r = 128$. These results establish that RMM can faithfully reconstruct the original compressed models while simultaneously delivering considerable storage efficiency.

A.3 RESULTS FOR 5 TASKS

For the experiments reported in Section 4, we adopted the RoBERTa-base model as the pretrained model and evaluated the model merging baselines on five GLUE tasks (QNLI, MRPC, SST-2, MNLI,

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	BOOLQ	Average
16	No merging	100%	91.74	71.54	95.30	84.19	88.14	71.47	83.73
	Task-Arithmetic	17%	50.34	37.16	58.49	34.43	63.30	41.87	47.60
	TIES	17%	49.79	36.75	61.24	35.57	63.55	40.67	47.93
	DARE	17%	49.95	37.68	56.08	34.51	63.39	42.78	47.40
	RMM ($p = 2$)	63%	88.12	44.00	93.58	73.06	85.63	59.69	74.01
	RMM ($p = 3$)	85%	91.58	49.04	95.64	80.31	87.52	65.50	78.27
32	No merging	100%	92.06	72.17	95.41	84.60	89.17	72.45	84.31
	Task-Arithmetic	17%	50.28	37.22	58.94	34.57	63.33	41.93	47.71
	TIES	17%	49.77	36.93	61.93	35.69	63.65	40.73	48.12
	DARE	17%	50.39	40.87	56.19	34.96	63.59	43.33	48.22
	RMM ($p = 2$)	56%	86.66	41.91	91.74	71.75	85.68	56.39	72.36
	RMM ($p = 3$)	76%	91.85	44.81	94.84	80.21	88.43	63.52	77.28
64	No merging	100%	92.26	72.12	95.41	84.64	90.07	72.84	84.56
	Task-Arithmetic	17%	50.28	37.28	59.06	34.56	63.34	42.08	47.77
	TIES	17%	49.75	37.04	61.81	35.84	63.74	40.76	48.16
	DARE	17%	50.16	40.52	58.83	35.59	63.35	42.54	48.50
	RMM ($p = 2$)	53%	83.87	42.72	89.45	69.76	85.70	53.43	70.82
	RMM ($p = 3$)	71%	92.07	41.97	94.50	79.70	88.73	61.96	76.49
128	No merging	100%	92.35	72.41	95.41	84.67	90.93	73.09	84.81
	Task-Arithmetic	17%	50.32	37.39	58.94	34.63	63.37	42.08	47.79
	TIES	17%	49.79	37.04	62.04	35.99	63.74	40.55	48.19
	DARE	17%	50.03	38.14	57.91	34.72	63.31	42.11	47.70
	RMM ($p = 2$)	52%	81.38	42.78	84.75	67.66	83.22	51.99	68.63
	RMM ($p = 3$)	69%	92.28	41.80	93.69	79.42	88.76	59.69	75.94

Table 3: Performance on the GLUE benchmark for the merging of six OPT-1.3b models compressed via PT-SVD across a range of ranks ($r = 16, 32, 64, 128$).

r	Method	Storage	MNIST	GTSRB	SVHN	Cars	EuroSAT	DTD	SUN397	Average
16	No merging	100%	99.61	97.22	97.06	69.20	99.67	75.21	72.72	87.24
	Task-Arithmetic	14%	47.19	28.29	38.64	56.27	57.52	41.38	62.92	47.46
	TIES	14%	60.14	34.52	49.03	58.55	61.70	43.67	62.87	52.93
	DARE	14%	53.80	31.00	44.96	57.60	56.93	43.99	62.42	50.10
	RMM ($p = 2$)	55%	99.16	88.52	94.89	66.61	97.44	60.64	67.86	82.16
	RMM ($p = 3$)	76%	99.58	95.65	96.54	68.30	98.93	67.18	69.50	85.10
32	No merging	100%	99.65	98.45	97.35	73.04	99.78	78.72	75.52	88.93
	Task-Arithmetic	14%	47.90	28.31	38.68	56.24	57.52	41.38	63.00	47.58
	TIES	14%	61.25	35.28	49.95	58.80	62.41	44.04	63.05	53.54
	DARE	14%	53.05	31.04	46.34	58.54	57.74	42.87	62.87	50.35
	RMM ($p = 2$)	49%	98.98	88.23	94.29	68.32	96.81	61.91	68.36	82.41
	RMM ($p = 3$)	67%	99.50	96.53	96.68	70.90	98.93	70.43	71.23	86.31
64	No merging	100%	99.67	98.68	97.41	75.76	99.85	79.15	77.73	89.75
	Task-Arithmetic	14%	47.99	28.30	38.71	56.21	57.59	41.38	63.22	47.63
	TIES	14%	61.83	35.93	50.28	59.08	63.07	44.15	63.17	53.93
	DARE	14%	54.78	32.79	45.05	58.23	61.30	42.87	62.77	51.11
	RMM ($p = 2$)	46%	98.46	84.43	92.83	69.36	95.59	62.39	69.77	81.83
	RMM ($p = 3$)	62%	99.45	96.14	96.33	72.84	98.56	72.07	73.85	87.03
128	No merging	100%	99.68	98.75	97.46	76.99	99.85	79.04	78.59	90.05
	Task-Arithmetic	14%	48.00	28.31	38.73	56.21	57.63	41.44	63.35	47.67
	TIES	14%	62.08	36.36	50.48	59.23	63.41	44.68	63.25	54.21
	DARE	14%	55.16	32.07	45.47	58.55	61.85	43.09	62.85	51.29
	RMM ($p = 2$)	44%	97.22	78.56	90.63	70.22	94.00	61.49	71.03	80.45
	RMM ($p = 3$)	59%	99.28	94.59	95.95	74.24	98.19	71.86	76.02	87.16

Table 4: Performance on vision tasks for the merging of seven ViT-B/32 models compressed via PT-SVD across a range of ranks ($r = 16, 32, 64, 128$).

QQP). Figures 1 and 2 in Section 4 summarize these results. For the sake of completeness, Table 5 corresponds to Figure 1 and reports the results for LoRA fine-tuned models, together with the improvements accruing from our method, whereas Table 6 corresponds to Figure 2 and reports the results for PT-SVD compressed models.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	Average
16	No merging	100%	61.38	83.48	94.50	83.90	87.56	82.16
	Task-Arithmetic	20%	50.61	33.51	73.85	34.23	64.58	51.36
	TIES	20%	50.65	33.51	75.57	35.83	66.97	52.51
	DARE	20%	50.61	33.51	79.36	34.70	63.40	52.32
	RMM ($p = 2$)	73%	59.64	68.64	93.00	82.68	86.89	78.17
32	No merging	100%	60.13	81.28	93.81	84.46	88.18	81.57
	Task-Arithmetic	20%	51.18	67.30	58.83	34.33	68.56	56.04
	TIES	20%	51.36	68.17	59.52	34.76	71.46	57.05
	DARE	20%	50.83	64.87	68.12	35.54	68.93	57.66
	RMM ($p = 2$)	66%	57.72	67.54	92.55	81.80	86.13	77.15
64	No merging	100%	59.62	84.12	94.15	85.20	88.75	82.37
	Task-Arithmetic	20%	54.74	68.93	80.28	35.70	36.82	55.29
	TIES	20%	55.26	68.93	78.78	36.76	36.82	55.31
	DARE	20%	54.71	68.81	74.54	38.72	36.82	54.72
	RMM ($p = 2$)	63%	59.82	67.48	91.51	83.21	84.41	77.29
128	No merging	100%	61.58	85.04	93.58	85.71	89.68	83.12
	Task-Arithmetic	20%	52.52	68.70	54.82	37.83	63.46	55.47
	TIES	20%	52.55	69.04	57.80	38.91	63.72	56.40
	DARE	20%	55.04	69.62	54.47	38.92	64.12	56.43
	RMM ($p = 2$)	62%	60.21	34.96	91.86	78.59	86.65	70.45

Table 5: Performance on the GLUE benchmark for the merging of five RoBERTa-base models compressed via LoRA across a range of ranks ($r = 16, 32, 64, 128$).

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	Average
16	No merging	100%	92.02	85.68	93.69	85.21	87.19	88.76
	Task-Arithmetic	20%	53.03	35.19	49.08	31.82	36.82	41.19
	TIES	20%	55.46	54.32	49.08	31.82	37.21	45.58
	DARE	20%	53.69	58.67	49.08	31.82	37.08	46.07
	RMM ($p = 2$)	73%	79.72	85.16	90.83	76.58	86.09	83.68
32	No merging	100%	92.31	85.86	94.15	86.38	88.62	89.46
	Task-Arithmetic	20%	53.23	35.59	49.08	31.82	36.82	41.31
	TIES	20%	55.78	55.19	49.08	31.82	37.62	45.90
	DARE	20%	55.89	33.74	49.20	31.82	36.85	41.50
	RMM ($p = 2$)	66%	63.87	82.78	91.28	77.12	87.05	80.42
64	No merging	100%	92.51	86.03	94.38	87.18	90.15	90.05
	Task-Arithmetic	20%	53.49	35.88	49.08	31.82	36.82	41.42
	TIES	20%	56.14	55.94	49.08	31.82	38.08	46.21
	DARE	20%	53.51	33.68	49.08	31.82	36.82	40.98
	RMM ($p = 2$)	63%	56.12	81.22	91.06	76.74	88.07	78.64
128	No merging	100%	92.68	86.14	94.50	87.33	91.05	90.34
	Task-Arithmetic	20%	53.60	36.06	49.08	31.82	36.82	41.48
	TIES	20%	56.40	56.35	49.08	31.82	38.61	46.45
	DARE	20%	50.52	34.96	49.08	31.82	36.92	40.66
	RMM ($p = 2$)	62%	54.11	79.19	90.60	74.49	87.46	77.17

Table 6: Performance on the GLUE benchmark for the merging of five RoBERTa-base models compressed via PT-SVD across a range of ranks ($r = 16, 32, 64, 128$).